

Sana Kang

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Education

University of Massachusetts Amherst Ph.D. in Computer Science (Advisor: Prof. Andrew Lan)	Amherst, MA, USA Incoming Fall 2026
Carnegie Mellon University , School of Computer Science Visiting Scholar (Advisors: Prof. Bhiksha Raj & Prof. Rita Singh)	Pittsburgh, PA, USA Jan – Jul 2025
KAIST (Korea Advanced Institute of Science and Technology) M.S. in Management Engineering (Advisor: Prof. Sunghyuk Park) B.S. in Computer Science and Business Technology Management	Daejeon, South Korea Feb 2024 – Feb 2026 Feb 2019 – Feb 2024

Publications & Patents

- [1] Nischal Ashok Kumar, Payu Wittawatolarn, **Sana Kang**, Marisa C. Peczuh, Blair Lehman, Ryan Baker, Caitlin Mills, Sherry Lachman, Ruochen Sun, Andrew Lan, “When Rubrics Change: Cross-Rubric Generalization for Critical Thinking Essay Scoring,” AI for Education Day at SIGKDD 2026, Jeju, Korea, 2026
- [2] **Sana Kang***, Myeongseok Gwon*, Su Young Kwon*, Jaewook Lee, Andrew Lan, Bhiksha Raj, Rita Singh, “PhoniTale: Phonologically Grounded Mnemonic Generation for Typologically Distant Language Pairs,” Proceedings of the Association for Computational Linguistics: EMNLP, Suzhou, China, 2025
- [3] Prithwiraj Choudhury, Do Yoon Kim, **Sana Kang**, “Beyond English-Centric AI: Strategic Frameworks for Developing Low-Resource Language Models,” Forthcoming in The Handbook of AI and Strategy (Csaszar & Jia, eds.; Edward Elgar Publishing)
- [4] Jaejung Roh, Hayder Albeit, Alistar Badinurmiyafar Abdulhusein, **Sana Kang**, Chanyoung Kim, “Method and System for Processing the Division of Blockchain-based Non-Fungible Token,” Korean Patent Application No. 10-2023-0161352 (filed Nov 20, 2023); Publication No. KR 10-2025-0074380 (published May 27, 2025).

Work in Progress

- [1] **Sana Kang**, Myeongseok Gwon, Jaewook Lee, Jinyeong Yi, Ha Eun Kim, Sejoong Kim, “SCRIBE: Structured Clinician-note Refinement with In-record Bounded Evidence.”
- [2] Myeongseok Gwon, **Sana Kang**, Minhyeong Lee, Sunghyuk Park, “LLMs Meet Match-Up Theory: A New Frontier in Influencer Recommendation.”

Research Experience

Discharge Summary Auto-Generation Using Language Models <i>Center for AI in Healthcare, Seoul National University Bundang Hospital</i> ◦ Researcher (Advisor: Dr. Jinyeong Yi, Center for AI in Healthcare) ◦ Leading the development of a LM-based discharge summary system targeting clinical deployment, including system design, IRB-approved data pipeline, and evaluation framework.	Feb 2026–Jun 2026 <i>Seongnam, South Korea</i>
Natural Language Processing with Phonetic Awareness <i>Language Technology Institute, Carnegie Mellon University</i> ◦ Independent Research (Advisor: Prof. Rita Singh and Prof. Bhiksha Raj) ◦ Formulated the research problem of cross-lingual phonological alignment and implemented LSTM-based phonological transformation and syllable modeling framework. (EMNLP’25) ◦ Initiated collaboration with Ph.D. Candidate Lee (UMass Amherst) by extending his baseline framework.	Mar 2025–Jun 2025 <i>Pittsburgh, PA, USA</i>
Causal Inference in Real Estate Markets <i>Tepper School of Business, Carnegie Mellon University</i> ◦ Research Assistant (Advisor: Prof. Minkyung Kim; with Prof. Meng Liu, WashU Olin) ◦ Investigated the causal impact of compensation transparency policies using multi-million data housing transactions, developing vectorized and memory-efficient preprocessing pipelines with polars . ◦ Resolved incomplete data issues for 400K+ listings by developing an efficient data acquisition pipeline, reducing reacquisition costs from \$300–\$400 to under \$3. ◦ Conducted state-level DiD estimation after policy-informed selection of key variables from 100+ raw features.	May 2025–Oct 2025 <i>Pittsburgh, PA, USA</i>
Healthcare AI Survey in Cardiovascular Prediction <i>Firenze Holdings, Inc.</i> ◦ Research Lead (Advisor: Dr. Eugene Hwang, Samsung Digital Health Team)	Dec 2024–May 2025 <i>Tokyo, Japan</i>

- Surveyed 70+ studies on cardiovascular prediction using EHR, ECG, imaging, and wearable data to benchmark ML approaches and propose multimodal and LLM-based strategies suited to the firm's constraints.

Blockchain-based Fractional NFT System

MIKES Lab, KAIST

Dec 2021–May 2022
Daejeon, South Korea

- Research Intern (**Advisor:** Dr. Hayder Albayati)
- Designed and implemented an ERC-1155 smart contract enabling algorithmic NFT fractionalization and verifiable ownership division, and developed an end-to-end DApp; patent filed (Publication No. 10-2025-0074380).
- Co-authored an engineering paper under the KAIST G-CORE industry collaboration program.

Work Experience

AI Engineer, Impact AI (Series Pre-A startup), South Korea

Jun 2025–Oct 2025

- HANKOOK TIRE: Designed production-scale RAG chatbot architecture; selected for development (Jan 2026).
- KYOBO LIFE INSURANCE: Led development and deployment of AI-based insurance recommendation service.
- ADOASIS: Developed an LLM-based marketing strategy recommendation model and full-stack MVP.

AI Engineer Intern, HdMedi, South Korea

Jul 2024–Sep 2024

- Led a small AI team to develop a Korean-specific prescription image parser and release an open-source library.

Machine Learning Intern, SweetSpot (Series B Startup), South Korea

Jun 2023–Feb 2024

- Built GraphSAGE-based model to predict sales and match franchise brands to specific commercial locations.

Research&Development Intern, UWS Blockchain Center, South Korea

Jun 2022–Dec 2022

- Developed hybrid wallet-based blockchain P2P trading system; contributed to patent (No. 10-2611819).
- Selected for Korea's Financial Regulatory Sandbox.

Data&Machine Learning Intern, LG Household & Healthcare, South Korea

Jul 2021–Aug 2021

- Developed price prediction model for Coca-Cola products and automated inventory management system.

Honors & Awards

Graduate Leadership Award, KAIST

Feb 2026

1st Prize, Global Data Convergence Talent Program Showcase, Ministry of Science and ICT

Nov 2025

Excellence Award, LLM Clinical Note Challenge, Seoul National University Bundang Hospital

Oct 2025

Academic Excellence Award, KAIST College of Business

Jan 2025

Selected Team Lead, AI Education Innovation Project (KRW 9.6M), KAIST

Sep 2024

Sponsor Prize (Polygon), Ethcon Korea, Ethereum Foundation

Sep 2023

Grand Prize, Startup Pitch Competition, Korea International Trade Association

Nov 2022

Merit Award, Best Leadership Presentation, KAIST Global Leadership Center

May 2022

Outstanding Graduate Award (Physics), Pusan National University Gifted Education Center

Jun 2021

Best Freshman Award, KAIST

May 2019

Silver Medal, Korean Young Physicists' Tournament, Korean Physical Society

Jan 2018

- Selected as a National Representative for the International Young Physicists' Tournament (IYPT).

Scholarships & Grants

Conference Scholarship (EMNLP'25), KAIST & IITP

Sep 2025

Sponsored AI Intensive Program at CMU (USD 41K), KAIST & IITP

Jan 2025–Jul 2025

Merit-based Full Tuition Scholarship (Graduate), KAIST College of Business

Feb 2024–Jul 2025

Merit-based KAIST Alumni Academic Scholarship (KRW 11.5M), KAIST

Apr 2022–Feb 2024

Merit-based Full Tuition Scholarship (Undergraduate), Korean government

Feb 2019–Feb 2024

Services and Leadership

External Reviewer, ICIS'25 (1 Full, 1 Short Paper), WITS'25 (2 Full papers)

2025

Student Vice President, Department of Management Engineering

2024

Organizer, KAIST Alumni Academic Scholarship Forum "Changing the World" @ InBody HQ

2023

Lead, Machine Learning Team, Google Developer Student Club (GDSC) @ KAIST

2023

- Led a year-long collaboration with a Series B startup on a location-based brand recommendation MVP.

Selected Contributor, National White Paper on University-Affiliated Science Gifted Education Centers

2021

Teaching and Mentoring

Teaching Assistant, BIZ591 IT Strategy and Management

Sep 2025–Dec 2025

- Developed and taught an end-to-end lightweight RAG Chatbot lab with a [Colab Notebook](#) and slides.

Teaching Assistant, BAF504 Investment Analysis

Sep 2024–Dec 2024

Mentor, AFE (All for Edu) Mentoring Program

2021–2022

- Supervised high school teams (Hampyeong Hakdari, Buyeo, Seochon Girls') on research projects.

Teaching Assistant, Physics Advanced Class, Pusan National University Gifted Education Center Jul 2018

- Assisted in lab experiments, supporting students' research and presentations during the final symposium.